ExoPhotoCurve 1.0.0 User manual

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Contents

ExoPhotoCurve — User Guide

ExoPhotoCurve is a program designed to create, inspect, correct and analyse photometric light curves, with a special focus on exoplanet transits.

The program can be used in two main ways:

1. **Analyse an existing light curve**, for example one exported from AstroImageJ, ExoClock or another photometry program.
2. **Create a light curve from calibrated and aligned FITS images**, using the integrated aperture photometry tool.

ExoPhotoCurve does not replace image calibration software. Bias, dark, flat correction, image alignment and plate solving must be done before using this program. ExoPhotoCurve works with already prepared light curves or with already calibrated and aligned images.

1. Starting the program

Start ExoPhotoCurve by running the main Python script or the Windows executable.

The main window contains:

- a control panel on the left;
- the light-curve plot on the right;
- several working tabs: **Data**, **Comp stars**, **Cleaning**, **Detrend**, **Transit**, **Statistics**, **Plot**, **Style** and **Binning**.

If you already have a light curve, you can load it directly.
If you want to create a light curve from FITS images, use:

```
Build light curve
```

2. Creating a light curve from FITS images

The **Build light curve** button opens the independent aperture photometry tool.

This tool creates a photometric table compatible with the main ExoPhotoCurve pipeline. The output table uses column names similar to those produced by AstroImageJ, so it can be loaded and analysed directly by the main program.

2.1 Image requirements

The FITS images must already be:

- bias/dark/flat calibrated;
- aligned;
- preferably all the same size;

- correctly ordered in time;
- equipped with reliable timing information in the FITS header.

The tool does not perform calibration, image alignment or plate solving.

2.2 Loading the FITS sequence

In the Aperture photometry window, select the folder containing your FITS images.

The program loads the sequence and displays a reference image. You can move through the sequence using:

```
arrow keys
previous/next buttons
Shift + mouse wheel
```

You can inspect the image using zoom and pan:

```
mouse wheel       = zoom
click + drag      = pan
Fit view          = show full image
```

2.3 Choosing the apertures

Set the aperture parameters:

```
aperture radius
sky inner radius
sky outer radius
search radius
```

The inner circle is the stellar aperture.
The outer annulus is used to estimate the sky background.

When you move the mouse over the image, the program shows the aperture at the current position. This helps you check whether the selected radius is appropriate for the stars.

The program also displays useful information about the region under the cursor, such as pixel value, peak value and mean signal inside the aperture.

2.4 Selecting the target and comparison stars

Before clicking on the image, choose which type of star you want to add:

```
Target
Comparison
Check
```

Then click near the star.

If **Snap click** is enabled, the program automatically moves the aperture to the local centroid of the nearest star within the search radius.

This means you do not need to click exactly at the centre of the star.

The selected apertures remain visible on the image.

You can delete or modify apertures before running the photometry.

2.5 Saving and loading apertures

You can save the aperture set to an external file.

This is useful if you want to repeat the photometry on the same sequence, or if you want to change only a few parameters without selecting all stars again.

When you reload an aperture set, the program restores the saved positions and displays them on the image.

The saved coordinates are reference positions. If **Recentre frames** is enabled, the program will recalculate the centroid of each star in every image during the photometry, within the selected search radius.

2.6 Recentering apertures during photometry

The option:

```
Recentre frames
```

makes the program search for the centroid of each star in every image.

This compensates for small alignment errors or small drifts of the field.

If the images are perfectly aligned, you can disable this option, but in most cases it is safer to keep it enabled.

2.7 Timing information in FITS images

Timing is one of the most important aspects of transit photometry.

Many FITS files store `DATE-OBS` or `JD_UTC` as the **start of the exposure**, not the mid-exposure time.

For 300-second exposures, the difference between exposure start and mid-exposure is:

```
150 seconds = 2.5 minutes
```

This can produce a visible offset in the measured O-C.

In the photometry window, choose:

```
FITS time ref:
- Exposure start
- Mid-exposure
- Exposure end
```

Then choose which time should be exported as the main JD_UTC column:

```
Main JD_UTC:
- Header time
- Mid-exposure corrected
```

Two safe workflows are:

```
Main JD_UTC = Header time
Transit tab = Exposure start
```

or:

```
Main JD_UTC = Mid-exposure corrected
Transit tab = Mid-exposure
```

The important point is consistency.

2.8 Running the photometry

When the apertures are ready, click:

```
Run photometry
```

The program measures the fluxes on the full FITS sequence and creates a photometric table.

The output file can contain columns such as:

```
JD_UTC
JD_UTC_header
JD_UTC_start
JD_UTC_mid
JD_UTC_end
AIRMASS
EXPTIME
FILTER
Source-Sky_T1
Source_Error_T1
Source-Sky_C2
Source_Error_C2
Source-Sky_C3
Source_Error_C3
rel_flux_T1
rel_flux_err_T1
```

The table can be saved and automatically loaded into the main ExoPhotoCurve window.

From this point on, the light curve is analysed using the standard ExoPhotoCurve pipeline.

3. Loading an existing light curve

If you already have a light curve exported from AstroImageJ, ExoClock or ExoPhotoCurve, you can load it directly into the main window.

Click:

```
Browse
Load table
```

The program reads the table and tries to identify the main columns automatically.

For AstroImageJ files, common columns are:

```
JD_UTC
JD_UTC_B
BJD_TDB
rel_flux_T1_dfn
rel_flux_err_T1_dfn
rel_flux_T1_dfn_model
rel_flux_T1_dfn_residual
```

If some columns are not recognised correctly, you can select them manually in the **Data** tab.

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4. Data tab

The **Data** tab is used to choose which columns are used for plotting and analysis.

The main columns are:

```
X / time
Light curve
Light-curve error
Model
Residuals
Residual errors
```

If your file does not contain a model or residuals, leave those fields as:

```
-- None --
```

The model can be generated later in the **Transit** tab.

.

5. Comp stars tab

The **Comp stars** tab is used to select the best comparison stars when the loaded file contains individual stellar fluxes.

This applies to complete AstroImageJ tables and to files generated by the ExoPhotoCurve aperture photometry tool.

The tab becomes active only if the program detects columns such as:

```
Source-Sky_T1
Source-Sky_C2
Source-Sky_C3
Source_Error_T1
Source_Error_C2
Source_Error_C3
```

If these columns are not present, the tab displays a warning and cannot be used.

5.1 Manual comparison-star selection

The list shows the available comparison stars.

You can activate or deactivate each star by clicking on its name.

When the subset changes, the light curve is recalculated automatically.

This allows you to reproduce manually the selection made in AstroImageJ, or to quickly check the effect of each comparison star on the final light curve.

5.2 Automatic comparison-star optimisation

The program can automatically search for the subset of comparison stars that produces the cleanest light curve.

You can use two workflows:

```
manual selection → automatic optimisation
automatic optimisation → manual refinement
```

In the first case, you manually exclude poor comparison stars and then let the optimiser work only on the remaining active stars.

In the second case, the program selects a subset automatically and you refine it manually afterwards.

The report shows the comparison between the initial curve and the optimised curve.

5.3 Sending the optimised curve to the pipeline

If:

```
Send optimised curve to Data/Transit
```

is enabled, the optimised curve is automatically selected as the main light curve in the **Data** tab.

You can then proceed with cleaning, detrending and transit fitting.

6. Cleaning tab

The **Cleaning** tab is used to remove problematic points.

There are two main methods:

```
automatic sigma clipping
manual point rejection
```

.

6.1 Sigma clipping

Sigma clipping identifies outliers based on their distance from the curve or from the residuals.

For exoplanet transits, use this function carefully, because overly aggressive clipping may remove real transit points.

A reasonable starting point is:

```
Target: Residuals
Sigma threshold: 3
Centre: Median
Scale: MAD
```

.

6.2 Manual point rejection

You can enable manual point editing.

When it is active, clicking near a point in the plot excludes or restores that point.

Manually rejected points are removed from the plot, statistics and transit fit, unless you explicitly choose to export rejected points.

.

7. Detrend tab

The **Detrend** tab is used to correct systematic trends in the light curve.

You can use one or more regressors at the same time, as in AstroImageJ.

Examples of possible regressors are:

```
JD_UTC / time
AIRMASS
FWHM
Sky background
X/Y centroid
Peak
Meridian flip
```

The detrending is applied multiplicatively:

```
corrected flux = observed flux / baseline
```

This is appropriate for relative-flux light curves.

7.1 Choosing regressors

By default, no regressor is active.

You can select regressors manually, or press **Suggested** to let the program enable likely useful regressors.

You can use several regressors together, for example:

```
JD_UTC + AIRMASS
```

or:

```
JD_UTC + AIRMASS + FWHM
```

7.2 Masking the expected transit during detrending

For exoplanet transits, it is usually recommended to use:

```
Mask expected transit
```

This tells the program not to use in-transit points when estimating the baseline.

This reduces the risk that the detrending will distort the transit depth.

7.3 Meridian flip

If the telescope performed a meridian flip during the sequence, you can use the dedicated correction.

A meridian flip may introduce a jump in flux level or a change in trend, because the field moves to a different region of the detector.

In the **Detrend** tab, enable:

```
Meridian flip
```

and enter only the decimal part of the flip time, as in AstroImageJ.

Example:

```
JD_UTC flip = 2461203.771
```

In the program, enter:

```
.771
```

You can press:

```
Update line
```

to display a vertical line in the plot at the flip position. This is only a visual check.

To actually apply the correction, click:

```
Run detrending
```

Available modes are:

```
Step only
Step + after-flip slope
```

The recommended mode is **Step only**, because it corrects a flux-level jump without adding too much freedom.

Use **Step + after-flip slope** only if there is a clear change in slope after the flip.

7.4 Removing detrending

If you want to return to the curve before detrending, click:

```
Clear detrending
```

This does not reset the full program. It removes only the detrending and returns to the source curve.

8. Transit modeling tab

The **Transit modeling** tab is used to model an exoplanet transit and produce a diagnostic report.

The typical workflow is:

```
1. Select the planet
2. Select the filter
3. Enter the exposure time
4. Select the input time system
5. Select whether the time stamps correspond to exposure start, mid-exposure
or exposure end
6. Enter the observatory coordinates
7. Click Run transit model
```

8.1 Planet catalogues

ExoPhotoCurve can use different catalogues:

```
NASA
ExoClock
custom catalogue
```

You can quickly switch between NASA and ExoClock using the dedicated buttons.

The ExoClock catalogue is often useful for updated ephemerides.
The NASA catalogue can provide geometric parameters useful for the physical transit model.

You can also load a custom catalogue, as long as it uses a compatible column format.

8.2 Automatic planet detection

When a file is loaded, ExoPhotoCurve tries to detect the planet from the filename.

Examples:

```
hats24b_20260606_lightcurve.txt → HATS-24b
wasp-15b_lightcurve.txt → WASP-15b
```

Always check that the selected planet is correct before running the fit.

8.3 Photometric filter and limb darkening

The selected filter is used by the physical model to choose approximate limb-darkening coefficients.

Changing the filter should not drastically change the transit, but it can slightly affect:

```
transit shape
ingress and egress
estimated Rp/Rs
residuals
```

The report shows the filter, limb-darkening law and coefficients used.

8.4 Time system and BJD_TDB correction

Transit timing requires careful time handling.

ExoPhotoCurve can convert `JD_UTC` times to `BJD_TDB` using:

```
target coordinates
observatory coordinates
exposure time
time-stamp reference
```

If the time column is `JD_UTC`, select:

```
Input time system = JD_UTC
```

If the times are already in `BJD_TDB`, select:

```
Input time system = BJD_TDB
```

The field:

```
Time stamp reference
```

defines whether each point represents:

```
Exposure start
Mid-exposure
Exposure end
```

This choice is very important. With 300-second exposures, confusing exposure start with mid-exposure produces a 2.5-minute timing error.

8.5 Fit baseline

In the **Transit** tab, you can choose a baseline model:

```
Constant
Linear
Quadratic
```

This baseline corrects small remaining trends during the transit fit.

If you have already applied a good detrending in the **Detrend** tab, **Constant** is often sufficient.

If residual trends remain, try **Linear** or **Quadratic**.

Use the quadratic baseline with caution if there is limited out-of-transit coverage.

8.6 Detrended or raw display

ExoPhotoCurve can display the transit fit in two ways:

```
Detrended flux
Raw flux with baseline
```

The recommended mode for final plots is:

```
Detrended flux
```

In this mode, the program shows the baseline-corrected light curve and clean transit models.

The mode:

```
Raw flux with baseline
```

is useful for checking how the baseline fits the original data.

8.7 Timing markers in the plot

You can show timing labels in the plot:

```
Predicted times
Calculated times
```

If **Predicted times** is enabled, the plot shows:

```
Predicted start
Prediction Tmid
Predicted end
```

If **Calculated times** is enabled, the plot shows:

```
Calculated start
Calculated Tmid
Calculated end
```

If both are enabled, the program also displays the O-C between the predicted and measured mid-transit times.

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8.8 Tmid ref

The field:

```
Tmid ref
```

allows you to manually enter a predicted mid-transit time in `BJD_TDB`.

Use this only if you have a predicted mid-time for the specific transit observed.

Do not enter an old ephemeris T0 here unless it corresponds to the observed transit.

.

9. Transit report

After **Run transit model**, ExoPhotoCurve produces a report containing:

```
planet
filter
exposure time
time system
BJD_TDB correction
target coordinates
observatory coordinates
catalogue used
T0 and period
predicted Tmid
observed Tmid
O-C
Rp/Rs
depth
duration
residual RMS
transit SNR
autocorrelation
diagnostic quality flags
```

The O-C is defined as:

```
O-C = observed Tmid - predicted Tmid
```

A positive O-C means that the transit occurred later than predicted.
A negative O-C means that it occurred earlier than predicted.

.

10. Reproducibility recipe

ExoPhotoCurve also saves a complete recipe of the analysis.

The recipe includes:

```
loaded file
catalogue used
selected columns
active comparison stars
rejected points
sigma clipping settings
detrending regressors
meridian flip settings
fit baseline
filter
time-stamp settings
observatory coordinates
main final parameters
```

You can save:

```
Save diag
```

to save the complete diagnostic report, or:

```
Save recipe
```

to save only the operational recipe.

This makes the analysis reproducible.

.

11. Saving the final light curve

The button:

```
Save curve
```

exports the light curve in its current analysis state.

The file can include:

```
JD_UTC
time_input
time_bjd_tdb
```

```
time
flux
flux_error
model
residual
baseline
point masks
```

To load the result into ExoClock, usually use:

```
JD_UTC
flux
flux_error
```

If you manually rejected points, ExoPhotoCurve saves only the points used in the analysis by default.

Enable:

```
Export rejected points
```

if you want to save rejected points as well, together with mask columns.

.

12. Binning

The **Binning** tab allows you to overplot binned points on the original light curve.

Binning is useful for visualising the general shape of the transit, but it should not replace the analysis of the original data points.

You can choose the number of points per bin and the style of the binned points.

.

13. Statistics

The **Statistics** tab computes basic quantities for the selected light curve, such as:

```
number of points
time span
median cadence
mean and median flux
RMS
amplitude
median error
RMS / median error
```

This is useful for quickly checking the quality of the light curve.

.

14. Reset and useful controls

ExoPhotoCurve includes several controls for reverting changes:

`Reset view/data`

restores the loaded table and removes temporary products, fits, binning, clipping and generated columns.

`Clear detrending`

removes only the detrending, while keeping the rest of the setup.

`Show transit fit on plot`

shows or hides the transit model in the plot.

15. Recommended workflow for an existing light curve

If you already have a light curve exported from AstroImageJ:

```
1. Load table
2. Check the columns in the Data tab
3. Use Comp stars if the file contains individual stellar fluxes
4. Remove problematic points in Cleaning
5. Apply detrending if needed
6. Select planet, filter, exposure time and time-stamp reference in Transit
7. Run transit model
8. Check plot, residuals, RMS and O-C
9. Save diag, Save recipe and Save curve
```

16. Recommended workflow starting from FITS images

If you want to do everything inside ExoPhotoCurve:

```
1. Build light curve
2. Load the FITS folder
3. Inspect image, zoom and stretch
4. Set aperture and sky annulus
5. Select target and comparison stars
6. Check FITS time ref and Main JD_UTC
7. Run photometry
8. Save the photometric table
9. Load the table into the main window
10. Use Comp stars to optimise the comparison subset
11. Apply manual cleaning or sigma clipping
12. Apply detrending with useful regressors
13. Run transit model
14. Save report, recipe and final curve
```

17. Practical advice

For reliable results:

- use well-calibrated images;
- avoid saturated stars;
- choose stable comparison stars that are not too faint;
- always check the time system;
- verify whether the FITS time is exposure start or mid-exposure;
- use meridian flip correction only when needed;
- avoid overly aggressive detrending if the transit is poorly covered;
- always inspect the residuals, not only the fitted curve;
- save the analysis recipe.

A good fit is not only a model passing through the data. A good analysis should have small residuals, no obvious trends, consistent timing and a reproducible procedure.

18. Program limitations

ExoPhotoCurve is designed for differential photometry, diagnostics and operational/preliminary exoplanet transit analysis.

It does not perform:

```
bias/dark/flat calibration
image alignment
plate solving
PSF photometry
full MCMC analysis
advanced physical modelling
```

The program is designed to be simple, transparent and practical. For final professional analyses, the results can be used as a basis or comparison, but they should be interpreted carefully.

19. Purpose of the program

The goal of ExoPhotoCurve is to let the user go from calibrated images or from a photometric table to a corrected, modelled and documented light curve.

The program helps the user control:

```
comparison stars
rejected points
detrending
timing
transit model
```

```
light-curve quality
final report
```

In this way, the analysis becomes more reproducible and less dependent on undocumented manual trial and error.